ELEMENT D

D1. PROCESS FOR GENERATING & COMPARING POSSIBLE SOLUTIONS

To begin developing a solution for redirecting A/C condensation away from the school walkway, I generated three different design ideas that approached the problem from different angles. The first idea involved creating a drip pan from a piece of sheet metal suspended under the A/C unit using jute twine. This metal sheet would act as a catch surface, guiding water into a rubber hose that drains into a 2.5-gallon bucket. The second idea was a simpler approach: placing a short piece of rubber hose directly under the A/C drip point so that the falling droplets would enter the hose and drain into a bucket. Although this design required minimal materials, it lacked stability and precision. The third idea involved forming a custom drip tray from a polycarbonate sheet. By heating and bending the edges upward, the tray could catch water and guide it toward a drainage hole connected to a rubber hose. This design also allowed for the addition of an inspection hole to check for blockages. After generating these ideas, I compared them using a weighted decision matrix based on the design requirements established earlier. Each requirement such as the ability to handle 2.5 gallons per day, durability, walkway safety, ease of installation, and inspection features was assigned a weight according to its importance. The sheet-metal drip pan scored 22 points, the hose-only design scored 16 points, and the polycarbonate drip tray scored the highest with 23 points. The polycarbonate design performed best because it met all high-priority requirements, offered durability, and provided a stable, controlled drainage system.

D2. DESIGN SOLUTION CHOICE WITH JUSTIFICATION

Based on the results of the decision matrix, the polycarbonate drip tray was selected as the final design solution. This design best satisfies the high-priority requirements, including durability, water resistance, and the ability to reliably redirect at least 2.5 gallons of water per day. Unlike the hose only design, the polycarbonate tray provides a wide catch surface that ensures no water misses the drainage system. Compared to the sheet-metal version, the polycarbonate tray is lighter, easier to shape, and resistant to rust or corrosion. It also allows for the integration of an inspection hole, which supports maintenance needs identified by stakeholders. Because it scored the highest in the weighted comparison and aligns with both research findings and stakeholder feedback, the polycarbonate drip tray is the most effective and practical solution.

D3. PLAN OF ACTION

The construction and testing of the final design followed a clear, chronological plan. During the first week, measurements of the A/C unit, drip point, and walkway distance were taken to ensure the tray would fit properly and drain water safely. All necessary materials including the polycarbonate sheet, rubber hose, jute cord, and super glue were gathered and recorded. In the second week, the polycarbonate sheet was cut to the correct dimensions and heated so the edges could be bent upward to form the tray. Once the tray was shaped, the rubber hose was securely attached to the drainage hole using super glue. After installation under the A/C unit, the system was tested for leaks or imperfections. The final test confirmed that the tray successfully redirected all water away from the walkway, keeping the area completely dry.

D4. DESIGN BLUEPRINT

The final design consists of a custom-shaped polycarbonate drip tray connected to a rubber drainage hose. The tray is made from a 26⅞-inch by 20-inch polycarbonate sheet. Using a heat gun, the edges of the sheet are bent upward at 90-degree angles to create a shallow basin capable of catching all condensation from the A/C unit. A small inspection hole is added to allow maintenance staff to check for blockages. A drainage hole is drilled at the lowest point of the tray, and a 4–5-foot rubber hose is attached using super glue. This hose directs water into a 2.5-gallon bucket placed safely away from the walkway. The materials required include one polycarbonate sheet ($20), one rubber hose ($8), jute cord ($9), and super glue ($4), for a total cost of approximately $31. This blueprint provides all necessary details so that anyone unfamiliar with the project could recreate the prototype.